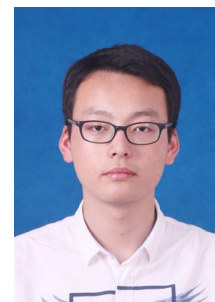


Yafei Chen (陈亚飞)



Personal Information

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Job: Software, Architect

Education

Shanghai Jiao Tong University	2017 ~ 2020	Master, Nuclear Technology Engineering
Shanghai Jiao Tong University	2013 ~ 2017	Bachelor, Nuclear Engineering

Skill

GPU architecture, CUDA, PTX,
LLM inference & training, DLRM, Graphics,
C++, python, Linux, bash

Experience

AI Performance Engineer Bytedance, 2024.7 ~ now

LLM Inference project

- Optimize E2E performance of Doubao and Deepseek-671B on heterogeneous HW
- Design GPU parallel algorithm, including TP, SP, EP and lower bit communication
- Develop faster page attention kernel, Moe dispatch and combine kernel
- Analysis profiling, fix bugs and improve serving throughput of PD separation framework
- Research latest quantification algorithms, like W4A8 and KIVI 4bit with page attention

LLM Training project

- Develop Faster Moe scatter & gather kernel to improve training performance
- Design online logits algorithm to reduce peak memory usage
- Develop LLM simulator to predict MFU and memory occupancy of different configurations

LRM (Large Recommendation Model) inference & training project

- Evaluation LLM&LRM performance of various new hardware
- Optimize LRM inference performance on TensorFlow framework through the development of deep fusion kernels and multi stream dispatch to reduce device bubbles
- Design embedding compression kernel to reduce H2D bandwidth bottleneck
- Develop hybrid embedding read and update algorithm, optimize all2all communication to improve LRM training performance
- Develop HSTU kernel for Generative Recommendation models

Resume

GPU Architect

Biren tech, 2022.7 ~ 2024.6

LLAMA-65B inference project

- Design TP, SP and multi GPU communication algorithm based on FasterTransformer
- Develop inference framework, AllReduce and AllGather kernels on BR104 GPU
- Fix bugs of BF16 precision to match golden result
- Analysis waveform to improve performance of key OP
- Build CI test plan, review code and manage release version

RTU (Ray Tracing Unit) model

- Design RTU architecture, including traversal algorithm and AS data format
- Develop module code, performance counter and unit test
- Profiling workloads of different scenarios and continuously optimize RTU architecture

GPU Software Engineer

Arm, 2020.4 ~ 2022.6

Develop UMD driver for Mali GPU

- Support new feature of GLES and Vulkan, like RayTracing extension
- Design nightly test, debug on model and deliver dump to RTL team
- Analysis performance of important benchmarks, including Aztec

Honor

SUPA programming contest 1st prize, 2023.2

Quarterly Award, 2021.4